

Legendrian Necks of the Positive Pseudosphere Are Convex Normal Lifts

Candidate full solution packet for arXiv:math/0610904

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Status. Candidate full solution, likely valid, pending human review. The theorem below answers the Legendrian-neck conjecture in Section 5.1 of Kuperberg’s paper for the smooth spacelike notion of neck used there.

1 Source problem

Let

$$H^+ = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^n : \langle x, y \rangle = 1\}.$$

Kuperberg equips the ambient space with the split bilinear form

$$\langle (\xi, \eta), (\xi', \eta') \rangle_Q = \frac{1}{2}(\langle \xi, \eta' \rangle + \langle \xi', \eta \rangle).$$

If a convex body $K \subset \mathbb{R}^n$ contains the origin in its interior, its positive normal lift is

$$K^+ = \{(x, y) \in \partial K \times \partial K^\circ : \langle x, y \rangle = 1\}.$$

On source PDF page 9, Section 5.1, the paper conjectures that every Legendrian neck of H^+ is K^+ for a convex body K [1]. The complete page is reproduced in the source-page evidence section.

The standard contact form on H^+ , up to a nonzero scalar and sign, is

$$\alpha = \frac{1}{2}(\langle y, dx \rangle - \langle x, dy \rangle).$$

A neck is, in particular, a compact connected spacelike submanifold of dimension $n - 1$ which is a section in Kuperberg’s topological sense. That section condition gives degree one for its projection to the sphere of y -directions.

2 Proof intuition

A point (x, y) of a convex normal lift records a boundary point x and its supporting covector y . The conjecture asks whether the contact condition and spacelikeness already force this convex interpretation.

They do. Legendrianity says that infinitesimal motion in x is orthogonal to y , so y behaves like a normal. The neck condition makes the normal direction a global parameter $u \in \mathbb{S}^{n-1}$. Writing $y = u/h(u)$, the contact equations recover the usual inverse Gauss parametrization $x = \nabla_{\mathbb{S}} h + hu$. The split metric on the neck becomes exactly the spherical support tensor $h^{-1}(\nabla_{\mathbb{S}}^2 h + hI)$. Its positivity is the strict-convexity condition. A geodesic ODE proves the global support inequality directly, so no local-to-global convexity theorem is needed.

3 The theorem

Theorem 1. *Let $n \geq 1$, and let $N \subset H^+$ be a smooth spacelike neck in the sense of Kuperberg. If N is Legendrian for $\ker \alpha$, then there is a unique smooth strictly convex body $K \subset \mathbb{R}^n$, with $0 \in \text{int } K$, such that $N = K^+$.*

Proof. For $n = 1$, a neck consists of one point (a, a^{-1}) on each of the two components of H^+ . If the first coordinates are $a_- < 0 < a_+$, it is exactly $[a_-, a_+]^+$, so the assertion is immediate. Assume henceforth that $n \geq 2$.

Write a tangent vector to N at (x, y) as $(\dot{x}, \dot{y})$. Since N lies in the hypersurface $\langle x, y \rangle = 1$,

$$\langle y, \dot{x} \rangle + \langle x, \dot{y} \rangle = 0. \quad (1)$$

Since N is Legendrian,

$$\langle y, \dot{x} \rangle - \langle x, \dot{y} \rangle = 0. \quad (2)$$

Adding and subtracting (1) and (2) gives

$$\langle y, \dot{x} \rangle = 0, \quad \langle x, \dot{y} \rangle = 0. \quad (3)$$

Consider the normal-direction map

$$\nu : N \longrightarrow \mathbb{S}^{n-1}, \quad \nu(x, y) = \frac{y}{|y|}.$$

It is a local diffeomorphism. Indeed, if $d\nu(\dot{x}, \dot{y}) = 0$, then $\dot{y} = ay$ for some scalar a . The second identity in (3) gives $a = \langle x, \dot{y} \rangle = 0$. Thus $\dot{y} = 0$, and the split squared norm of the tangent vector is

$$\langle (\dot{x}, \dot{y}), (\dot{x}, \dot{y}) \rangle_Q = \langle \dot{x}, \dot{y} \rangle = 0.$$

Spacelikeness then forces $(\dot{x}, \dot{y}) = 0$.

For completeness, the ambient projection

$$H^+ \longrightarrow \mathbb{S}^{n-1}, \quad (x, y) \longmapsto y/|y|$$

has contractible fibers: over u one may write $y = ru$ and $x = r^{-1}u + z$, with $r > 0$ and $z \perp u$. Hence it is a homotopy equivalence. The section condition in the definition of a neck implies that ν has degree of absolute value one. A local diffeomorphism from the compact connected neck to the sphere is a finite covering; its number of sheets is the absolute degree. Consequently ν is a diffeomorphism. This also covers the $n = 2$ case, where multiple coverings of the circle must explicitly be ruled out.

We may therefore parameterize N by $u \in \mathbb{S}^{n-1}$ and write

$$y(u) = \frac{u}{h(u)}, \quad h(u) > 0.$$

The equation $\langle x, y \rangle = 1$ says $\langle x(u), u \rangle = h(u)$. Moreover, (3) gives $\langle u, dx \rangle = 0$. Differentiating $h = \langle x, u \rangle$ in a tangent direction $w \in T_u \mathbb{S}^{n-1}$ therefore yields

$$dh(w) = \langle x, w \rangle.$$

The tangential component of x is thus $\nabla_{\mathbb{S}} h$, while its normal component is hu . Hence

$$x(u) = \nabla_{\mathbb{S}} h(u) + h(u)u. \quad (4)$$

Put

$$A_u = \nabla_{\mathbb{S}}^2 h(u) + h(u)I \quad \text{on } T_u \mathbb{S}^{n-1}.$$

Using the Euclidean derivative of the spherical gradient in (4), the two normal terms cancel and give

$$dx(w) = A_u w.$$

Also

$$dy(w) = h^{-1}w - h^{-2}dh(w)u.$$

Since $dx(w)$ is tangent to the sphere and A_u is self-adjoint, the metric induced on N is

$$g_N(w, z) = \frac{1}{2}(\langle dx(w), dy(z) \rangle + \langle dx(z), dy(w) \rangle) = h(u)^{-1} \langle A_u w, z \rangle. \quad (5)$$

Spacelikeness and $h > 0$ imply that A_u is positive definite for every u .

We now prove directly that h is a global support function. Define

$$K = \bigcap_{v \in \mathbb{S}^{n-1}} \{z \in \mathbb{R}^n : \langle z, v \rangle \leq h(v)\}.$$

Because h is positive and continuous, K is a compact convex body with the origin in its interior. Fix $u, v \in \mathbb{S}^{n-1}$ with $v \neq u$, and let $\gamma : [0, L] \rightarrow \mathbb{S}^{n-1}$ be a shortest unit-speed geodesic from u to v , where $0 < L \leq \pi$. Set

$$F(t) = h(\gamma(t)) - \langle x(u), \gamma(t) \rangle.$$

Equation (4) gives $F(0) = F'(0) = 0$. Since a Euclidean linear function restricted to a great circle satisfies $\ell'' + \ell = 0$, we have

$$F''(t) + F(t) = \langle A_{\gamma(t)} \dot{\gamma}(t), \dot{\gamma}(t) \rangle > 0.$$

Variation of constants gives

$$F(t) = \int_0^t \sin(t-s) \langle A_{\gamma(s)} \dot{\gamma}(s), \dot{\gamma}(s) \rangle ds.$$

For $0 < t \leq \pi$ this is strictly positive. In particular,

$$\langle x(u), v \rangle < h(v) \quad (v \neq u), \quad \langle x(u), u \rangle = h(u).$$

Thus $x(u) \in K$ and is the unique support point of K with outward normal u . It follows that h is the support function of K and $\partial K = \{x(u) : u \in \mathbb{S}^{n-1}\}$. The positive definiteness of A_u shows that this boundary is smooth and strictly convex.

Finally, $y(u) = u/h(u)$ lies in K° , because $\langle z, u \rangle \leq h(u)$ for every $z \in K$, and equality holds at $x(u)$. Hence $y(u) \in \partial K^\circ$ and $\langle x(u), y(u) \rangle = 1$. We have therefore shown

$$N = \{(x(u), y(u)) : u \in \mathbb{S}^{n-1}\} = K^+.$$

The support function $h(u) = 1/|y(u)|$ is determined by N , so K is unique. $\square$

4 Verification notes

- The sign of the contact form is immaterial. The level-set equation and Legendrian equation independently give the two vanishing pairings in (3).
- The only global topological input is already part of the source’s notion of a neck: its sphere-direction projection has degree one. This is needed in dimension two to exclude a multiple covering.
- Formula (5) checks the decisive sign: spacelike means $\nabla_{\mathbb{S}}^2 h + hI$ is positive, exactly the strict support-function tensor.
- The global support inequality is proved by the displayed geodesic ODE; no unquoted Hadamard or support-function realization theorem is used.
- The proof concerns smooth spacelike necks, exactly the class in the source conjecture. It does not claim the analogous statement for weak necks with null tangents.

5 Novelty check and limitations

On 21 July 2026, a bounded search covered the run registry, solution, attempt, and proof-gap indexes; the local source corpus; and web searches using the exact phrases “every Legendrian neck”, “Legendrian neck convex body”, and the source title together with “Legendrian”. The search returned the source paper and mirrors but no later paper answering this exact conjecture. This supports, but does not prove, novelty. The result should remain a candidate full solution until a specialist checks the contact convention and the degree-one consequence of the neck definition.

6 Source-page evidence

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This can be seen by supposing that $d\vec{x}$ and $d\vec{y}$ are orthogonal to each other and to both $\vec{x}$ and $\vec{y}$. Then the intersection window for $\vec{p}$ is a rectangle which is

$$\frac{\sin \beta - \alpha}{\sin \alpha} |d\vec{x}| \quad \text{by} \quad \frac{\sin \beta}{\sin \alpha} |d\vec{y}|.$$

(See Figure 4.) The factor $\vec{x} \wedge \vec{y}$ also produces a numerical factor of $\sin \alpha$ which must be cancelled. If we include variation of β , the region for $\vec{p}$ is a brick of thickness $d\beta$. To get an overall expectation, we must integrate the volume of this brick with respect to β and divide by $\text{Vol } S^3 = 2\pi^2$. Using

$$\int_{\alpha}^{\pi} (\sin \beta - \alpha)(\sin \beta) d\beta = \frac{(\pi - \alpha)(\cos \alpha) + (\sin \alpha)}{2},$$

the final answer is

$$\omega = \frac{(\pi - \alpha)(\cos \alpha) + (\sin \alpha)}{4\pi^2(\sin \alpha)^3} \vec{x} \wedge \vec{y} \wedge d\vec{x} \wedge d\vec{y}.$$

This formula agrees with equation 6.

5. OPEN PROBLEMS

5.1. Odds and ends

If the convex body K or its dual K° is not positively curved, then K^+ is a weakly spacelike section of H^+ , and K^- is a weakly timelike section of H^- . In other words, both will have some null tangents. In general if $N^\pm$ are weak necks of $H^\pm$ in $\mathbb{R}^{(a,b)}$, their join $N^+ * N^-$ might no longer be embedded except on a measure 0. For example, $K^+ * K^-$ is not embedded if K is a polytope.

Nonetheless, formula 2 is still the volume of a region $N^\diamond$. The necks are Lipschitz by virtue of being weakly spacelike and timelike, so the integrand is L^∞ . Moreover, the weak necks can be approximated by strictly spacelike and timelike necks, so the integrand is never negative; no volume is ever subtracted. We conjecture that in this generality, flat necks uniquely minimize the volume of $N^\diamond$. This is immediate from the proof of Theorem 1.4 for strict necks, but weak necks introduce new complications.

We do not know whether $K^\diamond$ is always convex when K is a centrally symmetric convex body. In light of the Mahler conjecture, we conjecture that $K^\diamond$ has the most volume when $K = C_n$. (We do not even know this when $n = 2$.) A related phenomenon is the fact that $K^\diamond = K \times K^\circ$ when K is a Hanner polytope [10]. We conjecture that they are the only symmetric convex bodies with this property, in light of the conjecture that they are the only convex bodies with minimum Mahler volume.

For two general necks $N^\pm$, $N^\diamond$ need not be convex and its volume is unbounded even in fixed dimensions. An upper bound may be possible if we suppose that $N^\diamond$ lies between H^+ and H^- . When $a = b = n$, then H^+ and H^- are contact manifolds, and the necks that come from convex bodies are Legendrian. We conjecture the converse (and it may not be hard to prove), that every Legendrian neck of H^+ is K^+ for a

convex body K . This extra geometric property of K^+ could be important for other problems in convex geometry.

Reference 13 discusses a different generalized bottleneck conjecture which is still open when $a, b > 2$. Instead of minimizing

$$\int_{N^+ \times N^-} \vec{x} \wedge \vec{y} \wedge d\vec{x}^{a-1} \wedge d\vec{y}^{b-1},$$

we can minimize

$$\int_{N_1^+ \times N_2^+} (\vec{x} \wedge d\vec{x}^{a-1}) \cdot (\vec{y} \wedge d\vec{y}^{b-1})$$

for two necks of H^+ . The dot product on $\Lambda^a(\mathbb{R}^{(a,b)})$ in this integral is induced from the one on $\mathbb{R}^{(a,b)}$. As before, we conjecture that the minimum occurs when the necks are flat and centered, and in this case coincident rather than orthogonal. When $N^+ = K^+$ for a symmetric convex body K , the two integrals are equal.

We can consider variations of the Mahler conjecture for complex and quaternionic convex bodies, *i.e.*, the unit balls of complex and quaternionic norms. The natural conjecture is that the unit balls of the complex and quaternionic ℓ^1 and ℓ^∞ norms have the least Mahler volume. (The ℓ^∞ ball is a polydisk, a Cartesian power of the Euclidean unit ball or disk in the complex numbers or quaternions.) Hanner polytopes also generalize analogously to Hanner bodies. If K is a complex or quaternionic convex body, then we can define not only the necks $K^\pm$, but more generally the neck K^z for any scalar $|z| = 1$. This extra geometry could lead to a variant of Theorem 1.4 and an improvement to Corollary 1.6.

5.2. Philosophy

We can attempt to apply the philosophy behind Theorem 1.4 and Corollary 1.6 to other problems in asymptotic convex geometry. Suppose that we want to bound some affinely invariant function $f(K)$ on convex bodies. Typically $f(K)$ is maximized (respectively, minimized) by ellipsoids; the question is to find a lower bound (respectively, an upper bound) to say that all convex bodies K are “not too far from round”. Our philosophy is to find another roundness statistic $g(K) \leq f(K)$ (respectively $g(K) \geq f(K)$) which is *minimized* (respectively *maximized*) when K is an ellipsoid. This yields a bound for the original statistic $f(K)$.

For example, the isotropic constant conjecture asserts that if $K \subset \mathbb{R}^n$ is a convex body, its isotropic constant $L(K)$ is bounded above by some constant C , independent of dimension [4]. (See also Reference [1].) The centrally symmetric case of the conjecture implies the general case [11]. The constant can be defined by minimizing an expectation:

$$L(K)^2 \stackrel{\text{def}}{=} \min_{K' = T(K)} \frac{E_{K'}[\vec{x} \cdot \vec{x}]}{n(\text{Vol } K')^{2/n}},$$

where the minimum is taken over all affine positions $K' = T(K)$, and the expectation is taken with respect to a random

Source PDF page 9. The target conjecture begins at the bottom of the left column and concludes at the top of the right column.

References

- [1] G. Kuperberg, *From the Mahler conjecture to Gauss linking integrals*, Geom. Funct. Anal. **18** (2008), 870–892; arXiv:math/0610904, DOI: 10.1007/s00039-008-0669-4.